# 3D Cluster Density Analysis

## Information Density in K-map Minimization

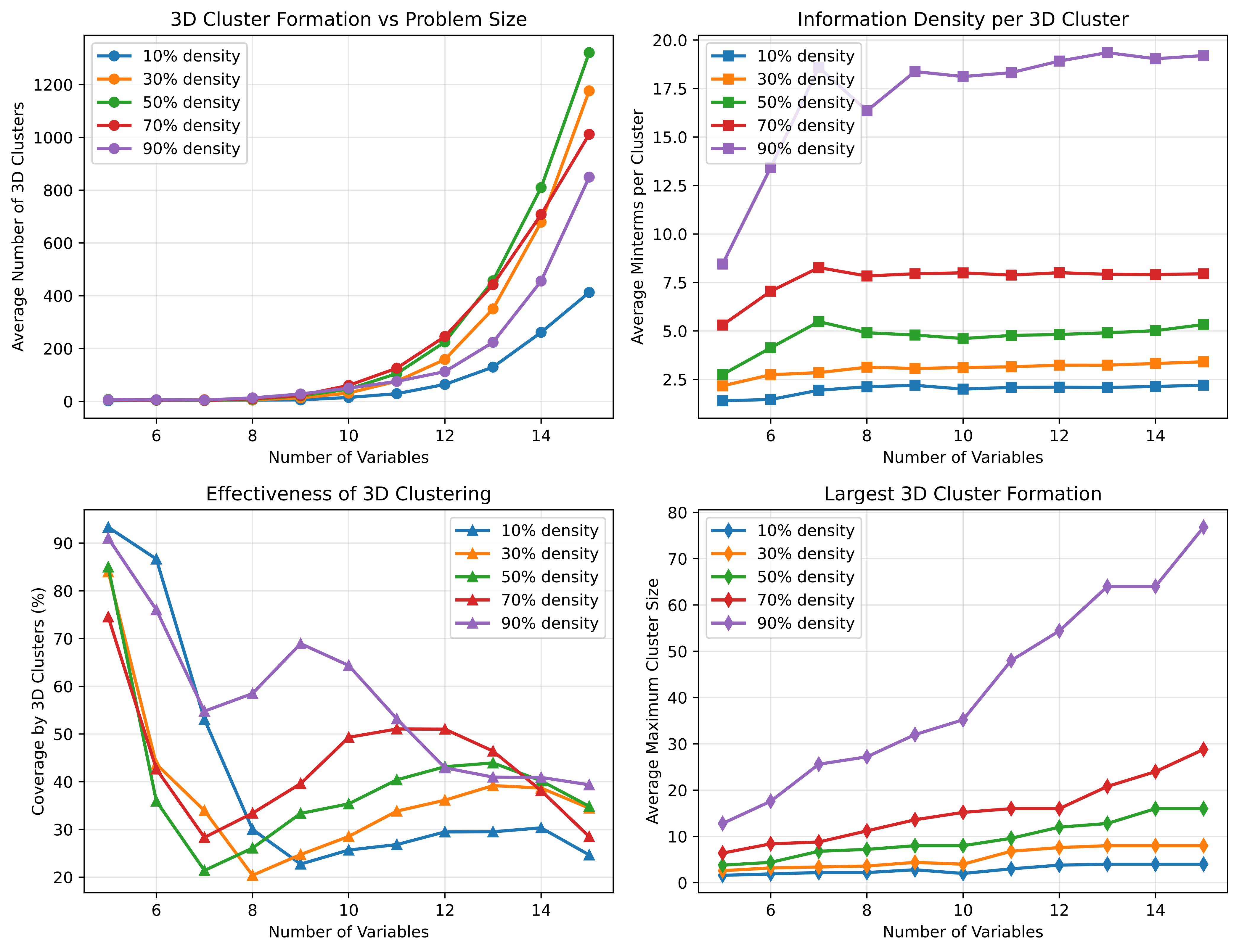
Analysis Date: 2026-01-19

Random Seed: 42

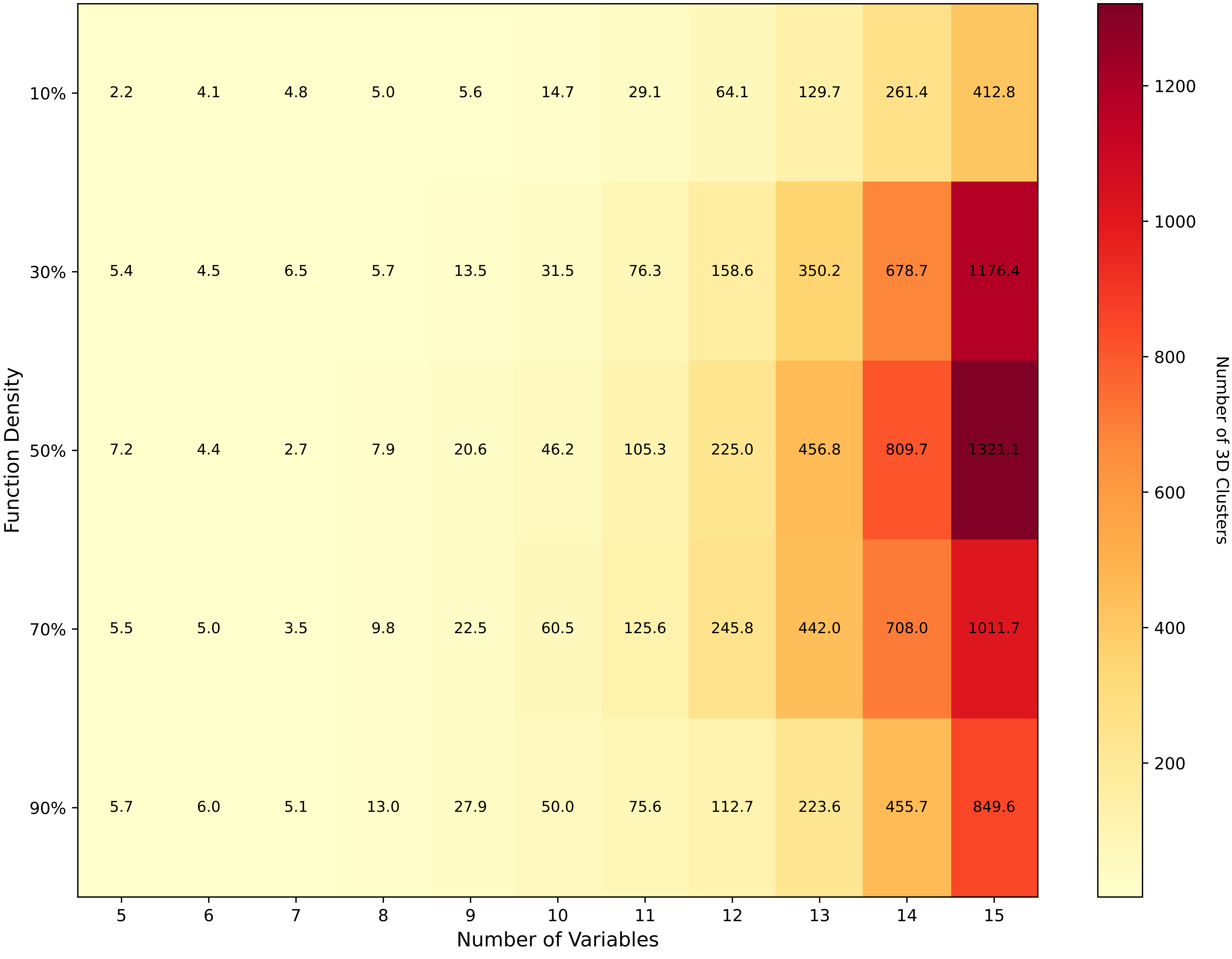
Total Test Cases: 550

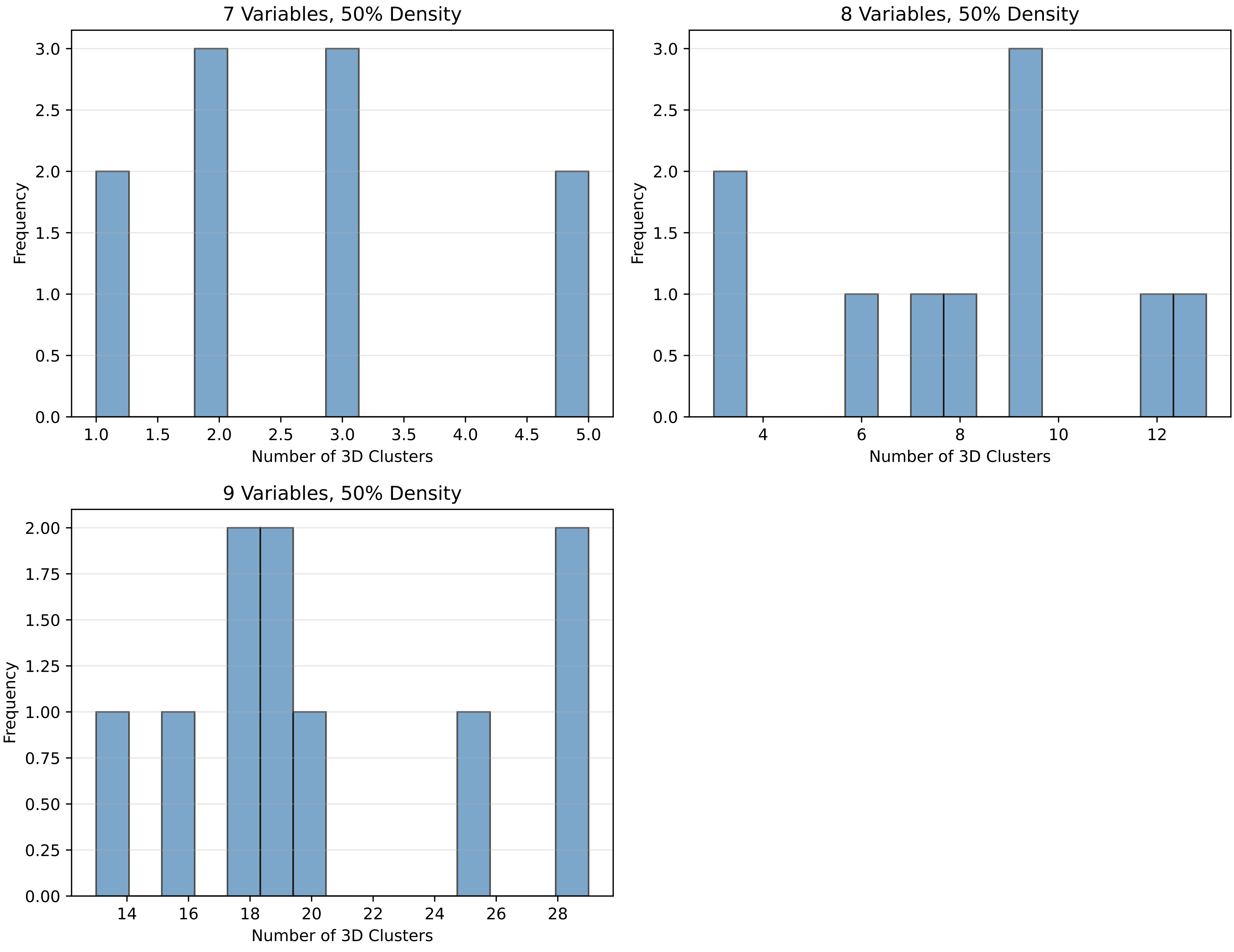
Variable Range: 5-15

*Geometric Clustering Behavior Analysis*


3D Cluster Formation vs Problem Size
Average Number of 3D Clusters
Number of Variables
Information Density per 3D Cluster
Average Minterms per Cluster
Number of Variables
Effectiveness of 3D Clustering
Coverage by 3D Clusters (%)
Number of Variables
Largest 3D Cluster Formation
Average Maximum Cluster Size
Number of Variables
10% density
30% density
50% density
70% density
90% density

**Average Number of 3D Clusters Formed\n(Heatmap)**





7 Variables, 50% Density
Frequency
Number of 3D Clusters
8 Variables, 50% Density
Frequency
Number of 3D Clusters
9 Variables, 50% Density
Frequency
Number of 3D Clusters

# Analysis Summary

## Key Findings

- Total tests conducted: 550
- Average 3D clusters per test: 198.41
- Average coverage by 3D clusters: 44.1%
- Variable range: 5 to 15
- Density levels tested: 5

## Observations

- *3D cluster formation increases with variable count*
- *Higher density functions tend to form more 3D clusters*
- *Average cluster size varies with function structure*
- *Coverage by 3D clusters is significant in most cases*

Report generated: 2026-01-19 16:38:56